

# Characterization of the Trans-Alfvénic Region Using Observations from Parker Solar Probe

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## ABSTRACT

Close to Earth the solar wind is usually super-Alfvénic, i.e. the speed of the solar wind is much larger than the Alfvén speed. However, in the lower coronal regions, the solar wind is mostly sub-Alfvénic. With the Parker Solar Probe (PSP) crossing the boundary between the sub- and super-Alfvénic flow, Bandyopadhyay et al. (2022) performed a turbulence characterization of the sub-Alfvénic solar wind with initial data from encounters 8 and 9. In this study, we re-examine the turbulence properties such as turbulence amplitude, anisotropy of the magnetic field variance, intermittency and switchback strength extending with PSP data for encounters 8–19. The later orbits probe lower altitudes and experience sub-Alfvénic conditions more frequently providing a greater statistical coverage to contrast sub- and super-Alfvénic solar wind. Also, by isolating the intervals where the solar wind speed is approximately equal to the Alfvén speed, we explore the transition in more detail. We show that the amplitude of the normalized magnetic field fluctuation is smaller for the sub-Alfvénic samples. While solar wind turbulence in general is shown to be anisotropic, the sub-Alfvénic samples are more anisotropic than the super-Alfvénic samples, in general. Further, we show that the sub- and super-Alfvénic samples do not show much distinction in terms of intermittency strength. Finally, consistent with prior results, we find no evidence for polarity reversing  $> 90^\circ$  switchbacks in the sub-Alfvénic solar wind.

**Keywords:** Space plasmas; Solar wind; Interplanetary magnetic fields; Alfvén waves

## 1. INTRODUCTION

Parker Solar Probe (PSP) (Fox et al. 2016; Raouafi et al. 2023) explores the solar atmosphere more closely than any previous *in situ* mission and is therefore able to observe processes potentially responsible for heating and accelerating the solar wind plasma. This already iconic dataset finally enables probing of the origin of the solar wind—a problem that has challenged the community

for more than half a century. It is now well understood that the plasma close to the sun is magnetically controlled and at least approximately corotates with the sun. At large distances the solar magnetic field can no longer control the plasma as the energy density in the flow greatly exceeds the thermal and magnetic energy densities. Between these limits are several interesting transitions where first, the flow energy density exceeds the magnetic energy density, and where second, the thermal pressure equals or exceeds the magnetic pressure. The first of these is commonly called the Alfvén critical point (Weber & Davis Jr 1967), now thought to be

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a more complex region with numerous such transitions (Chhiber et al. 2024). Exploring this “Alfvén transition region” in PSP data provides a unique opportunity to understand the differences in physical properties and processes that operate in the corona (the region below the transition) and the extended solar wind above.

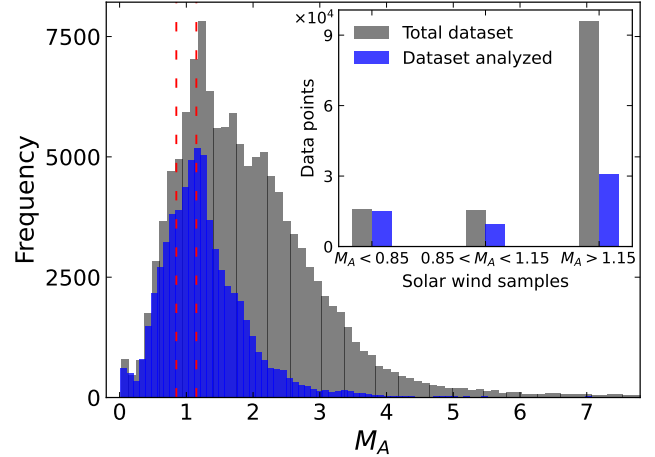
In regard to exploring the change in conditions across the trans-Alfvén transition, Bandyopadhyay et al. (2022) contrasted the properties of sub-Alfvénic solar wind as measured by PSP during the eighth (E8) and ninth (E9) solar encounters. The sub-Alfvénic datasets sampled in these earlier encounters provided a first opportunity to examine the sub-Alfvénic properties of the upper corona or sub-Alfvénic solar wind.

Here we extend that study to datasets from encounters E8 to E19. These later orbits probe lower altitudes and experience sub-Alfvénic conditions more frequently. With greater statistical coverage, we are able to better contrast the sub- and super-Alfvénic solar wind conditions. Furthermore, by isolating the intervals where the solar wind is trans-Alfvénic, (i.e. Alfvén Mach number  $M_A \simeq 1$ ), we explore the transition in more detail. Specifically, we analyze solar wind in categories defined as sub-Alfvénic  $M_A < 0.85$ , trans-Alfvénic  $0.85 \leq M_A \leq 1.15$ , and super-Alfvénic  $M_A > 1.15$  regimes and study their properties. Note that using a narrower range for the trans-Alfvénic regime, such as  $0.9 \leq M_A \leq 1.1$ , yields qualitatively similar results.

We note that other studies have also explored the trans-Alfvén transition (Zank et al. 2022; Zhao et al. 2022; Jiao et al. 2023) adopting strategies that are complementary to ours. The paper is organized along the lines of Bandyopadhyay et al. (2022): Section 2 describes the datasets employed. Section 3 provides details of the results on turbulence amplitudes (Bruno & Carbone 2013), variance anisotropy (Oughton et al. 2016), intermittency measured using partial variance of increment (PVI) methods (Greco et al. 2018), and occurrence rate of the switchback parameter (de Wit et al. 2020). Finally, in Section 4 we present our conclusions and discuss the implications.

## 2. DATA DURING ENCOUNTERS 8 – 19

Magnetic-field ( $\mathbf{B}$ ) data in the PSP measurements are obtained from the flux-gate magnetometer on the FIELDS instrument suite (Bale et al. 2016, 2019, 2020). The proton radial velocities ( $V_R$ ) are obtained from the Solar Probe ANalyzer for Ions (SPAN-I) on the SWEAP instrument suite (Kasper et al. 2016, 2019; Livi et al. 2020). The FIELDS QTN electron densities ( $N_e$ ) are computed from the quasi-thermal noise (QTN) spectrum (Romeo et al. 2023) measured by the



**Figure 1.** Comparison of distributions of the Mach number  $M_A$  for the analyzed dataset shown in Fig. 6 with the total datasets in encounters 8 – 19. The data set is divided into sub-Alfvénic ( $M_A < 0.85$ ), trans-Alfvénic ( $0.85 \leq M_A \leq 1.15$ ) and super-Alfvénic ( $M_A > 1.15$ ) samples by the red vertical dashed lines. Embedded: Comparison of the number of datasets in each category (sub-Alfvénic, trans-Alfvénic and super-Alfvénic).

Radio Frequency Spectrometer onboard PSP (Moncuquet et al. 2020). With these measurements, the local Alfvén speed is calculated as  $V_A = |\mathbf{B}| / \sqrt{\mu_0 m_p N_e}$ , where  $\mu_0$  is the magnetic permeability of a vacuum and  $m_p$  is the proton mass. Finally, the local Alfvén Mach number is estimated as  $M_A = V_R / V_A$  with a resolution of 60 s. The data is then divided into sub-Alfvénic, trans-Alfvénic and super-Alfvénic *regions* as intervals in which the sampled data satisfy the criteria  $M_A < 1$ ,  $0.85 \leq M_A \leq 1.15$ , and  $M_A > 1$  respectively, each lasting for a duration of 10 minutes. Further, all the calculations involving magnetic field uses a dataset of time resolution of 0.25 s, which are later resampled to map with the Alfvén Mach number  $M_A$ .

A graphical overview of the datasets collected in each encounter is given in the Appendix with the start and end times for each encounter listed in Table 2 (See Appendix). In Fig. 1, we compare the total datasets available in these encounters with the subset chosen for the analysis. For the purpose of this study, we use a total of 4 days of data for each encounter (enc) around the perihelion (two days on either side) to create a comparable sample size for both sub- and super-Alfvénic datasets and avoid statistical bias. Note that almost all of the sub-Alfvénic periods sampled by the PSP during these encounters have been included as shown in the bar chart embedded in Fig. 1. Many super-Alfvénic periods distant from the sun are excluded from our analysis.

### 3. RESULTS

#### 3.1. Turbulence Amplitude

First, we examine the amplitude of the magnetic field fluctuations. In Fig. 2a, we show the probability distribution function (PDF) of the turbulence amplitude, calculated as  $\delta B = \sqrt{\langle |\mathbf{B}(t) - \langle \mathbf{B} \rangle|^2 \rangle}$ , where  $\langle \dots \rangle$  is a time average, over some chosen time range, usually several correlation times. Throughout this paper, we choose averaging intervals of 10 minutes (on the order of few correlation times; see Parashar et al. (2020)) and evaluate the turbulence amplitude in each interval. Then, we separately accumulate the intervals classified as sub-Alfvénic, trans-Alfvénic or super-Alfvénic. In Fig 2a we show the frequency of occurrence of values of  $\delta B$  (in  $nT$ ), for super-Alfvénic and sub-Alfvénic intervals. We exclude the PDF of  $\delta B$  near unit Alfvén Mach number to clearly distinguish the two population. This result differs from that of Zank et al. (2022); Bandyopadhyay et al. (2022). In particular we cannot conclude that the sub-Alfvénic period have systematically lower  $\delta B$  as was reported in that earlier study. In fact there is a suggestion of a second population of sub-Alfvénic contributions with corresponding  $\text{Log}_{10}(\delta B) \sim 2.2$ . Inspection of the results from each encounter (not shown) indicates that this feature is mainly due to several encounters, such as E12, E17, E18, and E19. Most of the other encounters are more consistent with the conclusions in Bandyopadhyay et al. (2022). Fluctuations in the radial magnetic field ( $B_r$ ) are also small for sub-Alfvénic intervals and follow Bandyopadhyay et al. (2022). But the fluctuations in the normal ( $B_n$ ) and tangential ( $B_t$ ) magnetic fields are larger in the sub-Alfvénic intervals. Since the contribution to this secondary peak are mainly due to later encounters, we may hypothesize that larger amplitude transverse fluctuations are to be found in regions closer to the sun and perhaps inside the region of complex trans-Alfvénic transitions (Bandyopadhyay et al. 2022). Because of this potential complication, we examine now the turbulence amplitude parameter  $\delta B/B$ , where  $B$  is the average of the magnitude of the magnetic field over the interval i.e.  $B = \langle |B| \rangle_{\text{interval}}$ .

In Fig. 2b we show the frequency of occurrence of the normalized turbulence level (in logarithm), for the sub- and super-Alfvénic intervals. Clearly, the normalized turbulence amplitudes for the super-Alfvénic samples are larger relative to the sub-Alfvénic intervals. The most probable values of  $\delta B/B$  for the sub-Alfvénic, and super-Alfvénic wind are 0.20 and 0.38, respectively indicating stronger turbulence in super-Alfvénic samples compared with sub-Alfvénic samples. Similarly, the average value of the normalized turbulence amplitude for

the sub-Alfvénic solar wind is 0.18 with a standard deviation  $\sigma = 0.10$ . For the super-Alfvénic case, the average value of the normalized turbulence amplitude is 0.30 with  $\sigma = 0.17$ . While the trans-Alfvénic datasets are not included in these plots, the average and most probable values lie between the two regimes and are listed in Table 1 for comparison.

#### 3.2. Variance Anisotropy

In this subsection, we examine the variance anisotropy (Oughton et al. 2016; Parashar et al. 2016) of the magnetic field. Variance anisotropy measures the departure from equipartition of the magnetic field fluctuation energies in each cartesian component. For variance isotropy,  $\langle b_x^2 \rangle = \langle b_y^2 \rangle = \langle b_z^2 \rangle$ . Here, the mean magnetic field is excluded, usually by subtracting the average value  $\mathbf{B}_0$ . In a cartesian coordinate system, a measure of variance anisotropy is often defined as

$$A_b = \frac{\langle b_x^2 + b_y^2 \rangle}{\langle b_{\parallel}^2 \rangle}, \quad (1)$$

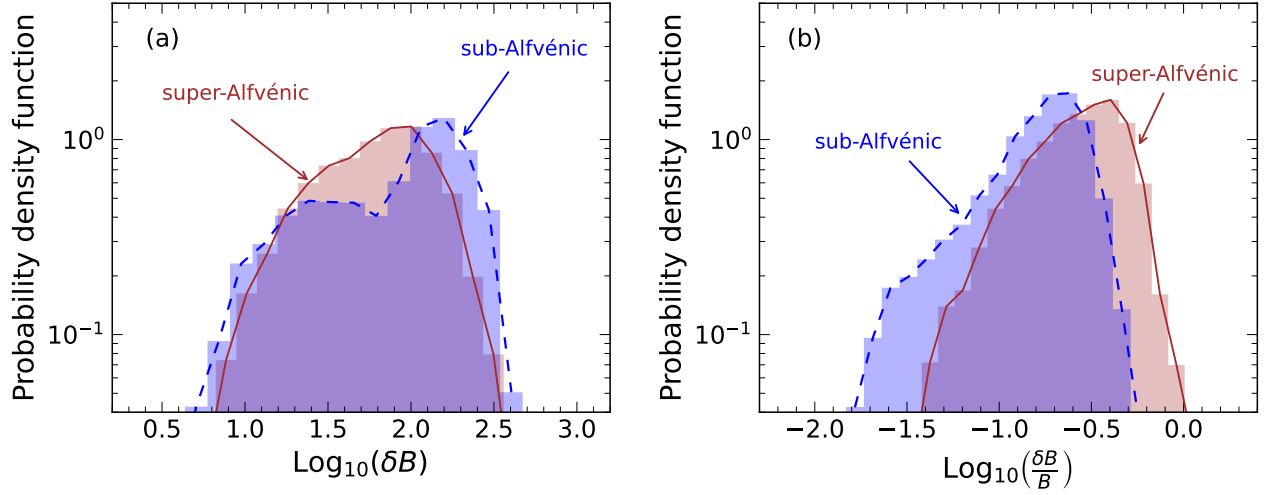
where  $\mathbf{B}_0$  is chosen to be along the z-axis and the magnetic fluctuations are  $\mathbf{b} = (b_x, b_y, b_{\parallel})$ . For an isotropic distribution of magnetic field components, the variance anisotropy takes on a value of  $A_b = 2$ .

The variance anisotropy  $A_b$  is calculated in each interval of 10-minute duration for the PSP datasets described in Section 2. The histograms of these magnetic field anisotropies for the sub-Alfvénic and super-Alfvénic periods are shown in Fig. 3. Again the intermediate trans-Alfvénic intervals are excluded from the histograms for clarity; with the average and most probable values shown in Table (1).

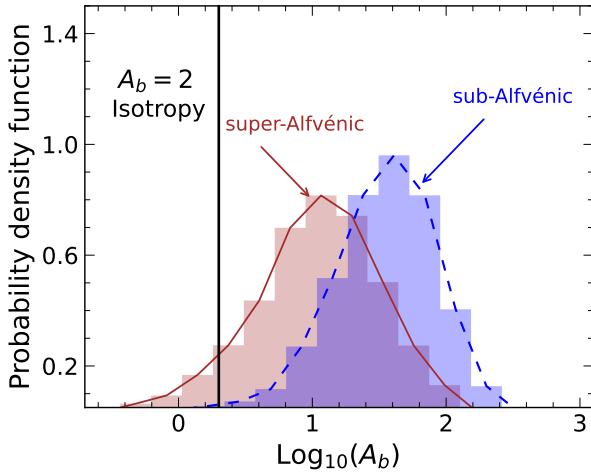
Clearly, all these three selected regimes of the solar wind are highly anisotropic compared to isotropy, for which one would expect  $A_b \sim 2$ , and  $\text{Log}_{10}(A_b) \simeq 0.3$ . However, the sub-Alfvénic solar wind periods display the strongest variance anisotropy of the three groups. The super-Alfvénic samples show the least variance anisotropy. From these distributions, one can also find the most probable value of the variance anisotropy  $\bar{A}_b = 56.23$  for the sub-Alfvénic case, and  $\bar{A}_b = 13.18$  for the super-Alfvénic case. Similarly, the average values of the variance anisotropy for the sub-Alfvénic case is  $\bar{A}_b = 51.09$  with a standard deviation of  $\sigma_{A_b} = 65.50$ . For the super-Alfvénic samples,  $\bar{A}_b = 20.16$  with  $\sigma_{A_b} = 30.33$ . The degree of variance anisotropy is clearly a distinguishing feature of the sub-Alfvénic corona as explored by PSP.

#### 3.3. Intermittency

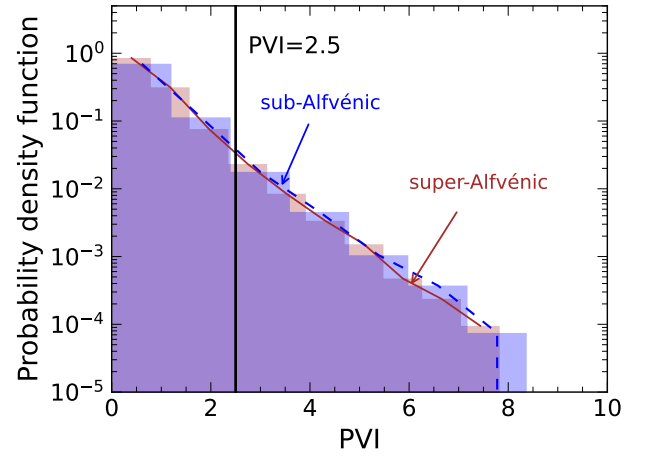
Intermittent signatures can be partially quantified by



**Figure 2.** Probability distribution function (PDF) of (a) the magnetic-field turbulence amplitude (in nT) and (b) the normalized magnetic-field turbulence amplitude in the sub-Alfvénic and super-Alfvénic solar wind intervals as observed by the PSP in encounters 8 – 19.



**Figure 3.** Probability distribution function of variance anisotropy in the sub-Alfvénic and super-Alfvénic solar wind intervals observed by PSP in encounters 8 – 19. The vertical solid (black) line indicates the value  $A_b = 2$ , which corresponds to isotropic distribution.



**Figure 4.** Probability distribution function of PVI in the sub-Alfvénic, and super-Alfvénic solar wind intervals observed by PSP in encounters 8 – 19. Values of  $PVI > 2.5$  represent the non-Gaussian and coherent structures such as current sheets.

the partial variance of increment (PVI) method, a measure that detects the occurrence of sharp gradients in quantities such as the magnetic field (Greco et al. 2008, 2018). The PVI of the magnetic field  $\mathbf{B}$  for a time lag of  $\tau$  at a time  $t$  is defined as

$$PVI_{t,\tau} = \frac{|\Delta \mathbf{B}(t, \tau)|}{\sqrt{\langle |\Delta \mathbf{B}(t, \tau)|^2 \rangle}}, \quad (2)$$

where the temporal increment of the magnetic field  $\mathbf{B}$  is defined as  $\Delta \mathbf{B}(t, \tau) = \mathbf{B}(t + \tau) - \mathbf{B}(t)$ .

Here PVI is computed using a moving average of 10 minutes, denoted by  $\langle \dots \rangle$ , with a time lag of  $\tau = 1s$ . Fig. 4 shows the distribution of values of PVI for the selected samples of sub- and super-Alfvénic solar wind. While the turbulence amplitude and variance anisotropy for the sub- and super-Alfvénic samples, as shown above, have distinctly different features, the PVI distributions separated in these two categories are very similar and display no distinct trends, consistent to the findings of Bandyopadhyay et al. (2022). The average value of PVI for both the sub- and super-Alfvénic cases is 0.75.

Likewise, the most probable values of PVI for both samples are approximately around 0.5 suggesting the close similarities between these samples with regard to PVI values.

### 3.4. Magnetic Switchbacks

Deflections of the magnetic field vector, including the phenomenon of “switchbacks”, may be quantified following de Wit et al. (2020) by the parameter

$$z = \frac{1}{2} (1 - \cos \alpha), \quad (3)$$

where

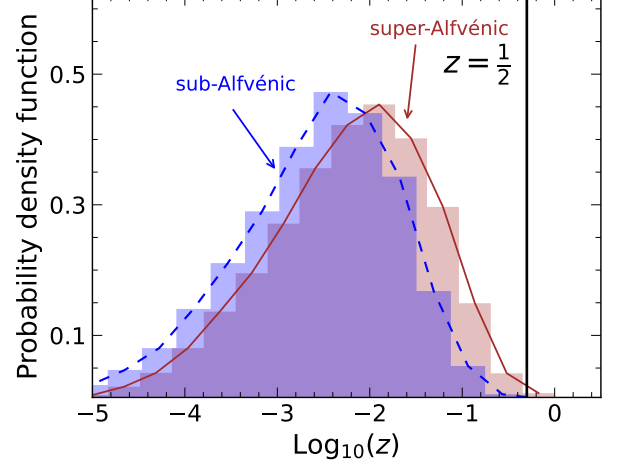
$$\cos \alpha = \frac{\mathbf{B} \cdot \langle \mathbf{B} \rangle}{|\mathbf{B}| |\langle \mathbf{B} \rangle|}$$

and the brackets denote a suitable local or regional average. Here, we again use an averaging interval of 10 minutes. The  $z$  variable admits values between 0 and 1. Values of  $z > 1/2$  indicate that the field is in a polarity-reversed state and lower values correspond to “background” magnetic polarity. In the present analysis, the term “switchback” refers exclusively to deflections that reverse the polarity i.e.,  $z > 1/2$ .

Figure 5 shows the histograms of the switchback parameter  $z$  (in logarithm) for the sub- and super-Alfvénic periods. Clearly, the magnetic deflections, as characterized by the  $z$  parameter, are smaller in the sub-Alfvénic periods than in the super-Alfvénic parts, and the deflections in the trans-Alfvénic samples lie between these. (As found by Bandyopadhyay et al. (2022); see also Jagarlamudi et al. (2023) for similar results.) The most probable values of the switchback parameter  $\tilde{z}$  for the sub-Alfvénic sample is 0.004, while for the Alfvénic and super-Alfvénic case, the most probable values  $\tilde{z}$  are 0.008 and 0.012 respectively. Similarly, the mean values of the switchback parameter  $\bar{z}$  for sub-Alfvénic solar wind is 0.012 with a standard deviation  $\sigma_z = 0.035$ . For the super-Alfvénic solar wind  $\bar{z} = 0.027$  with  $\sigma_z = 0.061$ .

## 4. DISCUSSIONS AND CONCLUSIONS

As PSP orbits descended more deeply into the solar atmosphere, the occurrence rate of sub-Alfvénic intervals has sharply increased, being less than 10% at 30-40  $R_s$  and increasing to >70% at 14  $R_s$  (Chhiber et al. 2024). Indeed, in more recent later orbits PSP has sampled extended periods of sub-Alfvénic solar wind. Comparing the sub-Alfvénic periods with the trans-Alfvénic and super-Alfvénic periods (see Table 1), we find that in the sub-Alfvénic solar wind: (i) the normalized turbulence amplitude decreases; (ii) anisotropy increases; (iii) intermittency as measured by PVI is almost unchanged; and (iv) the angular deflections of the magnetic field



**Figure 5.** Probability distribution function of switchback parameter ( $z$ ) in the sub-, and super-Alfvénic solar wind intervals observed by PSP in encounters 8 – 19. The vertical solid (black) line indicates the value  $z = 1/2$ , which corresponds to a marginal reversal of polarity of the magnetic field. Higher  $z$  values represent stronger switchback.

**Table 1.** Summary of the average (blue) and most probable (black) values of the several parameters described in the results section, for the sub-Alfvénic ( $M_A < 0.85$ ), trans-Alfvénic ( $0.85 \leq M_A \leq 1.15$ ), and super-Alfvénic ( $M_A > 1.15$ ) solar wind samples.

|                  |                                  | Sub-Alfvénic | Alfvénic | Super-Alfvénic |
|------------------|----------------------------------|--------------|----------|----------------|
| Turbulence       | $\frac{\delta B}{B}$             | 0.18         | 0.22     | 0.30           |
| Amplitude        | $\widetilde{\frac{\delta B}{B}}$ | 0.20         | 0.24     | 0.38           |
| Variance         | $\overline{A_b}$                 | 51.09        | 31.33    | 20.16          |
| Anisotropy       | $\tilde{A}_b$                    | 56.23        | 28.18    | 13.18          |
| Partial Variance | $\overline{PVI}$                 | 0.75         | 0.75     | 0.75           |
| of Increment     | $\widetilde{PVI}$                | 0.62         | 0.48     | 0.40           |
| Switchback       | $\bar{z}$                        | 0.012        | 0.015    | 0.027          |
| Parameter        | $\tilde{z}$                      | 0.004        | 0.008    | 0.012          |

are on average weakening, indicating the lack of switchbacks (Ruffolo et al. 2020). Overall the behavior is consistent with the initial study by Bandyopadhyay et al. (2022) except that in the present study we have much better statistical coverage of the sub-Alfvénic samples. Once again we can see that the sub-Alfvénic wind more closely resembles properties that one might associate with coronal conditions. Often the corona is modeled based on properties expected for low beta, highly anisotropic plasma, which may be dominated by incompressive fluctuations such as Alfvénic fluctuations (or



waves). A typical coronal model of this type is based on Reduced Magnetohydrodynamics (RMHD) as employed in, e.g., Einaudi & Velli (1999); Gomez et al. (2000); Oughton et al. (2001). In contrast, the super-Alfvénic solar wind is much less anisotropic and is typically modeled in the fluid regime by compressible MHD models without the reduction in dimensionality leading to RMHD. In this regard it has been shown that the validity of the RMHD approximation is particularly sensitive to the presence of component variance in the direction parallel to the regional mean magnetic field (Dmitruk et al. 2005). From this perspective it appears that, as it approaches the sun, PSP is sampling not only a complex transition from super-Alfvénic to sub-Alfvénic wind (Chhiber et al. 2024), but also a transition from a large plasma beta, more fully three dimensional compressible MHD plasma (at scales much larger than the ion inertial scale) to a plasma better described as highly anisotropic, less compressible and described by Reduced MHD. We note that the anisotropy measure shown in Fig. 3 is essentially the reciprocal of the quantity sometimes called magnetic compressibility (Kiyani et al. 2012); its behavior is consistent with greater compressibility in the super-Alfvénic regime. Other signatures of this transition, such as the presence of co-rotation as a signature of magnetic dominance in the corona, are anticipated (Weber & Davis Jr 1967; Kasper et al. 2019; Chhiber et al. 2025).

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## 5. APPENDIX

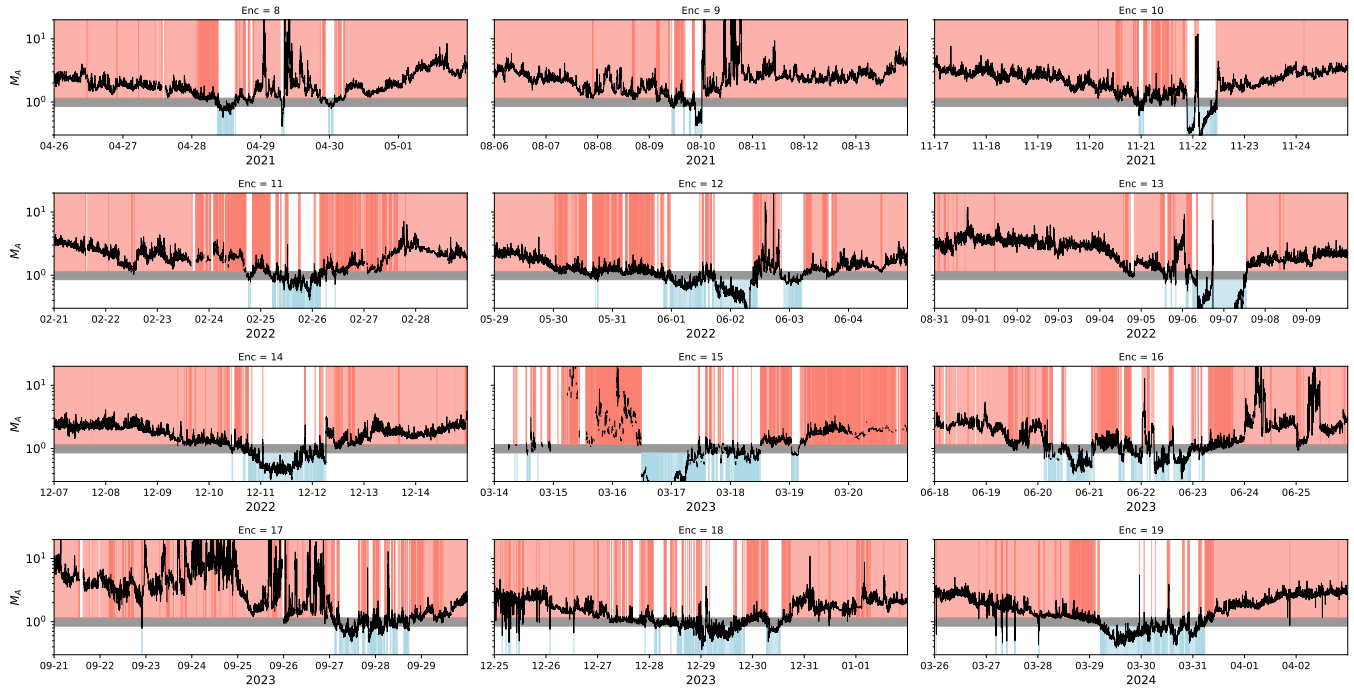
Here we provide an overview of the PSP datasets from encounters 8–19. In Fig. 6, the datasets are shown characterized based on Alfvén Mach number. Sub- and super-Alfvénic data samples are represented by the shaded regions with lighter shades of blue and red respectively, with an additional gray region separating the **Table 2**. Start and end times of the 4 days of dataset (centered at the perihelion) analyzed for Parker Solar Probe (PSP) encounters 8–19.

| Enc | Start (UTC)         | End (UTC)           |
|-----|---------------------|---------------------|
| 8   | 2021-04-27 06:00:30 | 2021-05-01 06:00:30 |
| 9   | 2021-08-07 18:00:30 | 2021-08-11 18:00:30 |
| 10  | 2021-11-19 09:00:30 | 2021-11-23 09:00:30 |
| 11  | 2022-02-23 12:00:30 | 2022-02-27 12:00:30 |
| 12  | 2022-05-31 00:00:30 | 2022-06-04 00:00:30 |
| 13  | 2022-09-04 06:00:30 | 2022-09-08 06:00:30 |
| 14  | 2022-12-09 12:00:30 | 2022-12-13 12:00:30 |
| 15  | 2023-03-15 21:00:30 | 2023-03-19 21:00:30 |
| 16  | 2023-06-20 00:00:30 | 2023-06-24 00:00:30 |
| 17  | 2023-09-26 00:00:30 | 2023-09-30 00:00:30 |
| 18  | 2023-12-27 00:00:30 | 2023-12-31 00:00:30 |
| 19  | 2024-03-28 00:00:30 | 2024-04-01 00:00:30 |

trans-Alfvénic dataset ( $0.85 \leq M_A \leq 1.15$ ). The dates of the intervals from each encounter used in this study are given in Table 2.

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**Figure 6.** An overview of the Alfvén Mach number  $M_A = V_R/V_A$  where  $V_A$  is the Alfvén speed, as observed by the Parker Solar Probe (PSP) for encounters 8 – 19. Sub- and super-Alfvénic intervals are shown by the shaded regions colored light blue and light red respectively. In addition, shaded gray region extending  $0.85 \leq M_A \leq 1.15$  represents what we have termed the “trans-Alfvénic dataset” isolated to contrast the behavior between sub- and super-Alfvénic solar wind.

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